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1I / 01 / 2241720263

Code :

```java
Codeium: Refactor | Explain | Generate Javadoc
int[] toArray(){
    int [] data = new int[size()]; // It creates an empty array.
    Node2P tmp = head;
    int index = 0;
    while (tmp != null) { // It traverses the DLL using a loop, retrieves the data from each node, and adds it to the array.
     data[index++] = tmp.data;
     tmp = tmp.next;
    }
    return data; // Finally, it returns the populated array.
}

//soal 2
Codeium: Refactor | Explain
DLL sublist(int start, int end){
    DLL sublist = new DLL();
    if (start < 0 || end >= size() || start > end) { // It checks if the given start and end positions are valid within the range of the DLL.
        System.out.println("Invalid sublist range!!");
        return sublist;
    }
    Node2P tmp = head;
    int index = 0;
    while (tmp != null && index <= end) {
        if (index >= start) { // If the positions are valid, it creates a new DLL object, subList, and copies the elements from the DLL's start position
        //to the end position into subList.
            sublist.addLast(tmp.data);
        }
        tmp = tmp.next;
        index++;
    }
    return sublist; // Finally, it returns the subList.
}
```

```java
//soal 3
Codeium: Refactor | Explain
void addAll(DLL list){
    Node2P tmp = list.head; // It adds all the elements from the given list DLL to the tmp DLL.
    while (tmp!=null){ // It uses a loop to iterate over each element in list and calls the addLast() method to add it to the current DLL.
        addLast(tmp.data);
        tmp = tmp.next;
    }
}

//soal 4
Codeium: Refactor | Explain
boolean containsAll(DLL list){
    Node2P tmp = list.head;
    while (tmp != null) { // It uses a loop to iterate over each element in list and checks its existence in the current DLL using the contains()
        // helper method.
        if (!contains(tmp.data)) { // It checks if all the elements from the given list DLL exist in the current DLL.
            return false;
        }
        tmp = tmp.next;
    }
    return true;
    // If any element is not found, it returns false; otherwise, it returns true.
}

Codeium: Refactor | Explain | Generate Javadoc
boolean contains(int value) {
    Node2P tmp = head;
    while (tmp != null) {
        if (tmp.data == value) {
            return true;
        }
        tmp = tmp.next;
    }
    return false;
}
```

```java
//soal 5
Codeium: Refactor | Explain
void removeAll(DLL list){
    // It removes all the elements from the current DLL that also exist in the given list DLL.
    Node2P tmp = list.head;
    while (tmp != null) { // It uses a loop to iterate over each element in list and calls the deleteByData() method to remove the corresponding
     // elements from the current DLL.
     deleteByData(tmp.data);
     tmp = tmp.next;
    }
}
```

Result :

```
┌──(zharsuke㉿asus-vivobook)-[~/…/College/Semester_2/Data_Structure_and_Algorithm_Practicum/uas]
└─$  /usr/bin/env /usr/lib/jvm/java-17-openjdk-amd64/bin/java -XX:+ShowCodeDetailsInExceptionMessa
f1da50744e032ea38f728380efd6/redhat.java/jdt_ws/uas_267c1de3/bin DLLMain
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
Initial data :
10-100-34-20-200-75

Display data from toArray() function :
 10 100 34 20 200 75

Display data from sublist() function :
100-34-20

Display data from addAll() function :
10-100-34-20-200-75-212-212-212

Display data from containsAll() function :
true

Display data from removeAll() function :
10-20-200-75

┌──(zharsuke㉿asus-vivobook)-[~/…/College/Semester_2/Data_Structure_and_Algorithm_Practicum/uas]
└─$
```